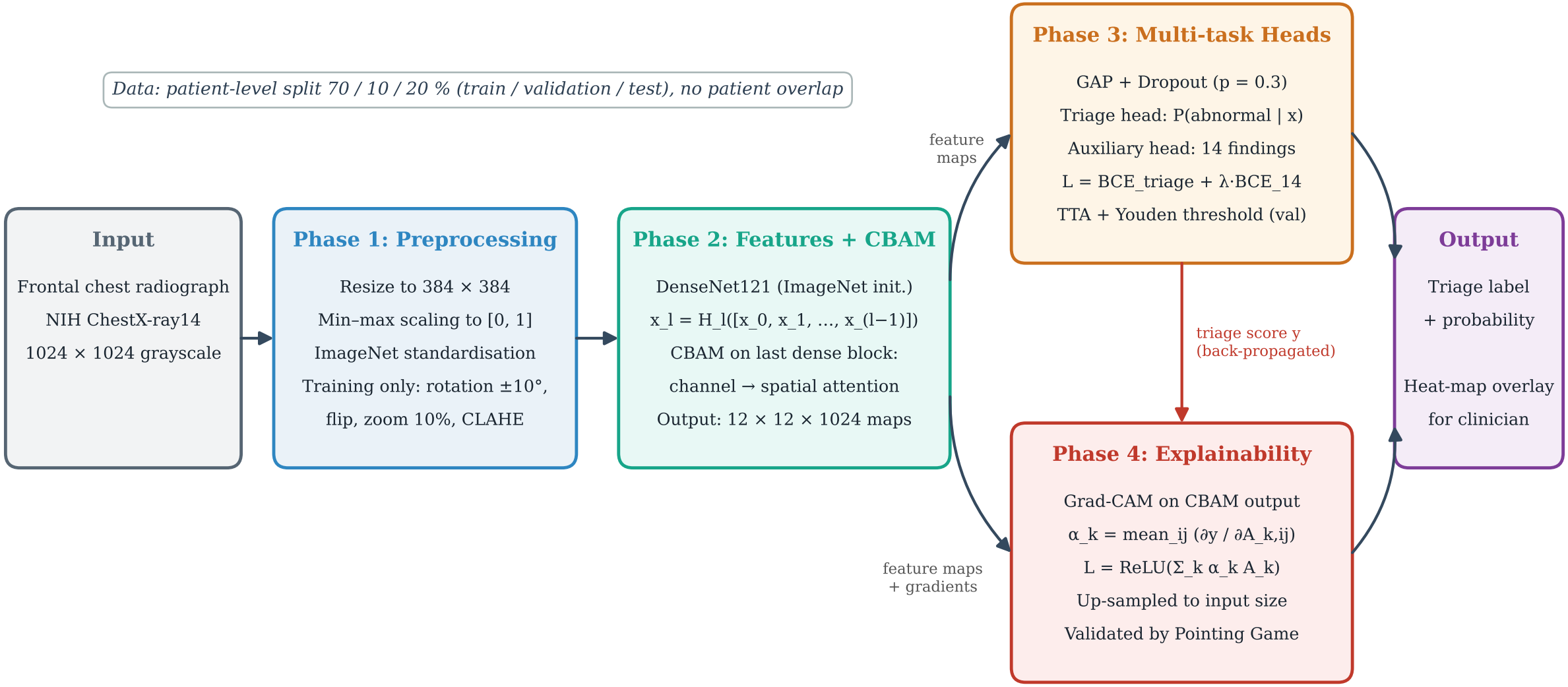
Data: patient-level split 70 / 10 / 20 % (train / validation / test), no patient overlap
Input
Frontal chest radiograph
NIH ChestX-ray14
1024 × 1024 grayscale
Phase 1: Preprocessing
Resize to 384 × 384
Min–max scaling to [0, 1]
ImageNet standardisation
Training only: rotation ±10°,
flip, zoom 10%, CLAHE
Phase 2: Features + CBAM
DenseNet121 (ImageNet init.)
x_l = H_l([x_0, x_1, ..., x_(l−1)])
CBAM on last dense block:
channel → spatial attention
Output: 12 × 12 × 1024 maps
feature
maps
Phase 3: Multi-task Heads
GAP + Dropout (p = 0.3)
Triage head: P(abnormal | x)
Auxiliary head: 14 findings
L = BCE_triage + λ·BCE_14
TTA + Youden threshold (val)
triage score y
(back-propagated)
feature maps
+ gradients
Phase 4: Explainability
Grad-CAM on CBAM output
α_k = mean_ij (∂y / ∂A_k,ij)
L = ReLU(Σ_k α_k A_k)
Up-sampled to input size
Validated by Pointing Game
Output
Triage label
+ probability
Heat-map overlay
for clinician